

Product Requirement Document (PRD)

Product: Low-Code/No-Code (LC/NC) Hospital Web Application

Version: 1.0

Prepared For: Healthcare SaaS Platform Stakeholders

Date: 25 September 2025

1. Executive Summary

Hospitals currently rely on developers to implement customizations in their web applications, which causes delays and complexity. Doctors, nurses, and staff want to self-configure forms, workflows, and validations without developer dependency.

This PRD defines a Low-Code/No-Code Hospital Web Application that enables self-service customization while ensuring compliance (DPDP, ABDM/NDHM), multi-tenancy, and scalability across multiple hospitals.

2. Problem Statement

1. All customization requests currently go to developers, increasing turnaround time.
2. Different personas (doctors, nurses, staff) demand personalized configurations.
3. Increasing customizations lead to complexity and rollback delays.
4. Doctors want a DIY solution (self-service) with guardrails.

3. Goals & Objectives

- Enable hospitals to configure forms, fields, rules, workflows without developers.
- Provide a microservice-based architecture that is scalable and modular.
- Ensure data isolation per tenant (multi-tenancy).
- Provide fine-grained access control (RBAC).
- Ensure compliance with DPDP Act & ABDM/NDHM.
- Guarantee auditability, integrity, and security of medical data.

4. User Personas

- **Doctor:** Wants to add/remove form fields, validations, and triggers.
- **Nurse:** Wants quick workflows for vitals, alerts, and escalations.
- **Receptionist/Clerk:** Wants easy data entry and validation for insurance, guardian details.
- **Administrator:** Wants tenant-level customization, access control, and compliance.
- **Compliance Officer:** Needs audit logs, consent management, and regulatory reporting.

5. User Stories / Use Cases

1. **Emergency Triage:** Priority field required. If Critical → show Injury Type & alert trauma team.
2. **Diabetes Visit:** Auto-calculate BMI from height/weight. If HbA1c ≥ 9 → book educator session + notify endocrinologist.
3. **Pediatrics:** If age < 18 → guardian details + birth certificate upload mandatory. Workflow → reminder task if missing file.
4. **Insurance/TPA:** If Insurance = Yes → require Policy & Insurer. Workflow → call TPA system for authorization.
5. **Hypertension:** If BP > 140/90 → re-check required. If still high → referral to cardiology.

6. Functional Requirements

- Form Configurator: Add/hide fields, validations, versioning & rollback.
- Workflow Engine: Define workflows, retries for failures.
- Runtime Engine: Apply configurations dynamically.
- Submission Gateway: Validate, deduplicate, and save records.
- Core Medical Records: Store in ABDM/FHIR format.
- File Management: Secure uploads, virus scan.
- Multi-Tenancy & RBAC: Tenant isolation, granular permissions.
- Compliance & Consent: DPDP compliant, ABDM integration, audit trail.

7. Non-Functional Requirements

- Scalability: Handle 100+ hospitals and millions of records.
- Performance: Sub-second response for form load.
- Security: JWT, RBAC, consent checks.
- Reliability: Retry workflows, safe submissions.

- Compliance: DPDP, ABDM/NDHM.

8. System & Data Flow

Microservices: User Management, Configurator, Runtime Engine, Submission Gateway, Core Service, Document Service, Workflow Orchestrator.

Communication: REST/gRPC, Kafka for async.

Data Isolation: Tenant-aware models.

Audit Logging: Immutable, hash-chained.

9. Success Metrics / KPIs

- 80% reduction in customization turnaround time.
- 70% of changes done by staff without IT.
- 99.9% uptime.
- 100% compliance pass rate.
- Rollback < 5 minutes.

10. Risks & Assumptions

- Risk: Misconfigured rules block workflows → Mitigation: Versioning + rollback.
- Risk: Staff adoption → Mitigation: Simplified UI + training.
- Assumption: Hospitals have IT infra & internet.

11. Future Enhancements

- AI-assisted rule recommendations.
- Natural language rule builder.
- Offline-first sync (Service Worker + IndexedDB).
- Insurance system integrations.